

Two MacMahon counterexamples to Conjecture 425

gaochengzhecpu (AI-assisted submission)

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Source and interpretation. Conjecture 00000000425 (original statement and pinned commit in `SOURCE.md`) asserts that the coefficient sequence in the diagonal specialization of the boxed plane partition generating function is always log-concave. Write $F_n(q)$ for the standard volume generating polynomial of the $n \times n \times n$ box, so the three box dimensions are specialized to the diagonal. We disprove both log-concavity of the numerical coefficients of a fixed F_n and coefficientwise log-concavity of the polynomial sequence (F_n) . Both arguments are formalized in the accompanying Lean file.

Definitions. For the box of side two, a plane partition is an array $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with entries in $\{0, 1, 2\}$, weakly decreasing along both rows and columns: $a \geq b$, $a \geq c$, $b \geq d$, and $c \geq d$. Its volume is $a + b + c + d$. Let c_k count such arrays of volume k . Log-concavity requires $c_k^2 \geq c_{k-1}c_{k+1}$ at every interior index.

Counterexample and proof. There is exactly one array of volume zero, namely the zero array, and exactly one of volume one, with a single 1 in the top-left entry. The three arrays of volume two are

$$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

These are exhaustive: either one entry equals 2, which by monotonicity must be the top-left entry, or two entries equal 1, giving exactly the two displayed order ideals. Consequently

$$c_0 = 1, \quad c_1 = 1, \quad c_2 = 3, \quad c_1^2 = 1 < 3 = c_0c_2.$$

Thus the asserted universal log-concavity is false. It is unnecessary to consider the additional equality characterization after this first clause has failed.

Independent polynomial check. The complete polynomial, obtained both by direct enumeration and by the MacMahon product, is

$$\prod_{i,j,k=1}^2 \frac{1 - q^{i+j+k-1}}{1 - q^{i+j+k-2}} = 1 + q + 3q^2 + 3q^3 + 4q^4 + 3q^5 + 3q^6 + q^7 + q^8.$$

The accompanying Python verifier independently computes the product as an exact integer formal power series and enumerates the arrays.

Direct formal connection to the MacMahon product. Write its numerator and denominator as $N(q)$ and $D(q)$. For the cubic box of side two, the exact triple products give

$$D(q) = 1 - q - 3q^2 + O(q^3), \quad N(q) = 1 - q^2 + O(q^3).$$

Since $D(0) = 1$, formal division defines a coefficient sequence f with $D(q)F(q) = N(q)$. Comparing the first three coefficients gives $f_0 = 1$, $f_1 - f_0 = 0$, and $f_2 - f_1 - 3f_0 = -1$; thus again $(f_0, f_1, f_2) = (1, 1, 3)$. This directly refutes log-concavity for the product specified in the conjecture, without using the counting formula as an unproved assumption.

The self-contained Lean 4.19.0 file `lean/Main.lean` formalizes both routes. It verifies the complete, duplicate-free enumeration of the 81 arrays and the precise plane-partition predicate. It also defines integer formal series, Cauchy multiplication, and the actual three nested MacMahon products. The terminating recursion `formalQuotient` constructs every coefficient; `formalQuotient_spec` proves $DF = N$ at every index whenever $D(0) = 1$. The theorem `macmahon_product_counterexample` therefore refutes log-concavity for an explicitly constructed infinite coefficient sequence, not for a predicate whose existence is merely assumed. Its first three coefficients are also proved equal to the direct enumeration above. All calculations use kernel proofs, with no placeholders or custom axioms.

Coefficientwise log-concavity across box sizes. For a sequence of polynomials, coefficientwise log-concavity requires every coefficient of $F_n(q)^2 - F_{n-1}(q)F_{n+1}(q)$ to be nonnegative. This reading fails at $n = 2$ as well. The completely filled box is the unique term of maximum volume, so F_n is monic of degree n^3 . Consequently F_2^2 has degree 16 while F_1F_3 is monic of degree 28, and

$$[q^{28}](F_2(q)^2 - F_1(q)F_3(q)) = 0 - 1 = -1.$$

In particular, the claimed nonnegativity fails before any equality characterization is considered.

Formal verification of the cross-size argument. The Lean proof does not assume the preceding degree or monicity facts. It constructs three actual infinite formal quotients from the exact MacMahon numerators and denominators for side lengths 1, 2, and 3. Their full equations $D_n F_n = N_n$ hold at every index by `formalQuotient_spec`; in each case $D_n(0) = 1$ is verified.

To calculate efficiently, the file represents the first 29 coefficients by finite integer lists. The theorem `prefixProduct_correct` proves for arbitrary lists of factors that the truncated product agrees with the infinite Cauchy product at every index below the cutoff. Successive kernel checks compute the actual MacMahon factors, rather than assume their product or a claimed polynomial table. Three further kernel checks verify $D_n C_n = N_n$ through degree 28 for the displayed coefficient lists C_n . The theorem `equation_prefix_unique`, proved by strong induction, then identifies those lists with the first 29 coefficients of any true MacMahon quotient. This uniqueness step is valid because $D_n(0) = 1$.

Finally `macmahon_cross_size_coefficient` applies to arbitrary series satisfying the three MacMahon equations and proves the displayed coefficient to be -1 . The theorem `concrete_cross_size_coefficient` specializes it to the constructed quotients, so there is no existence assumption left open. The theorem `macmahon_diagonal_coefficientwise_counterexample` negates the universal coefficientwise assertion using its $n = 2$ instance. Thus both readings described at the start have full formal disproofs.

The independent script `verify.py` concerns only Conjecture 425. It enumerates the 2×2 arrays, computes the three exact MacMahon series through degree 28, compares them with the Lean coefficient tables, and checks the cross-size convolution. Only the standard Lean foundational axioms occur; no additional axiom or external decision oracle is used.